

Full Length Article

Relationship between postural control and muscle activity during a handstand in young and adult gymnasts



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ABSTRACT

The aim of this study was to evaluate the relationship between muscle activity and inter-muscle contributions patterns and postural control during a handstand. Additionally, outcomes were compared between young and adult gymnasts (mean $\pm$ SD: 13.9 $\pm$ 0.7 and 23 $\pm$ 3 years respectively). Participants performed three trials of a 10 s handstand on a force platform with simultaneous EMG signal recording at the upper and lower limbs. Adult gymnasts demonstrated significantly better postural control in each studied variable. The wrist flexors demonstrated the highest relative mean (60%) and peak (200%) EMG activity of all muscle groups studied. Wrist flexor activity was also highly correlated with postural control variables in both groups. The trapezius descendens and wrist flexor muscles demonstrated the highest contribution (20–26% and 25.5–28% respectively), followed by anterior deltoid (15–18%) and triceps brachii (13–16%) in both groups. The young gymnast group demonstrated significantly greater mean relative muscle activity at the triceps brachii, biceps brachii and rectus femoris compared with the adult group (88% ($p = 0.023$), 150% ($p = 0.003$) and 75% ($p = 0.039$) respectively). To conclude, despite comparable inter-muscle contributions during a handstand, young and adult gymnasts demonstrated a different relationship between muscle activity and postural control variables.

1. Introduction

An upright standing position is natural for human beings. It is learned during the early stages of development and allows individuals to perform upper limb tasks while walking (Bairstow & Laszlo, 1981; Winter, 1995). However, some individuals also develop the ability to stand on their upper limbs for the purposes of entertainment and sport.

An untrained person is incapable of maintaining steady, static balance during a handstand. It is a difficult task which requires a high degree of muscle strength and endurance to hold the full weight of the body as well as the cardiovascular endurance required to withstand the inverted position (Shvartz, 1968). A performer must also develop a great sense of balance to maintain a static position, an ability which depends on proprioceptive, vestibular and visual sensors (Gautier, Thouvarecq, & Chollet, 2007; Sobera, Siedlecka, Piestrak, Sojka-Krawiec, & Graczykowska, 2007). Maintaining balance in an inverted position is difficult due to: 1) the smaller base of

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support, 2) higher center of mass than in upright standing, and 3) the lesser strength of the wrist joints compared to the ankles (Gautier, Marin, Leroy, & Thouwarecq, 2009; Kerwin & Trewartha, 2001; Prassas, 1988). The ability to control the center of mass relative to the base of support is one of the aims of postural control, and requires the integration of vestibular, somatosensory and visual information (Horak & Macpherson, 2011; Massion, 1994).

Despite the fact that the upper limbs are not adapted to support bodyweight, upper limb kinematics during a handstand is similar to lower limb kinematics during upright standing (Slobounov & Newell, 1996). The wrist joints act similarly to the ankle joints, and the elbow and the shoulder joints act similarly to the knee and hip joints respectively (Gautier et al., 2007; Kerwin & Trewartha, 2001). The maintenance of balance is initially controlled by the wrists and then shoulder with hip strategies being adopted when postural sway becomes too great or in response to a perturbation (Kerwin & Trewartha, 2001; Yeadon & Trewartha, 2003).

Previous researches have investigated a number of aspects of postural control during a handstand including center of pressure (CoP) sway (Asseman, Caron, & Crémieux, 2004; Croix, Chollet, & Thouwarecq, 2010; Sobera et al., 2007), dynamics analysis (Gautier et al., 2009; Kerwin & Trewartha, 2001; Slobounov & Newell, 1996), ecological theory (Gautier et al., 2007), and limited electromyographic (EMG) analyses (Clement & Rezzette, 1985; Yeadon, Rosamond, & Hiley, 2012). However, there is lack of extensive research regarding neuromuscular contribution during a handstand and its relationship to postural control in terms of posturography.

Although previous research has provided insight into how particular joints contribute to postural control during a handstand (Blenkinsop, Pain, & Hiley, 2017; Kerwin & Trewartha, 2001; Yeadon & Trewartha, 2003) these studies did not investigate the involvement of individual muscles. Investigation of postural sway during handstands has been primarily conducted using time-dependent analyses such as cross-correlation of EMG feedback time delays (Blenkinsop, Pain, & Hiley, 2016) and regression modelling of torque joint (Yeadon & Trewartha, 2003). As the ability to maintain the center of gravity in a stable position is dependent upon constant position correction via muscle activity (Borg, Finell, Hakala, & Herrala, 2007; Houdijk, Brown, & van Dieën, 2015), a correlation between the magnitude of muscle activity (amplitude) and postural control variables should be observed during a handstand. Muscle activity preventing the loss of balance briefly utilizes additional fast twitch fibers to enable a quick reaction (Daly, 2010; Johnson & Woollacott, 2011). Therefore, maximal CoP sway and displacement velocity should be more strongly correlated with peak EMG signal than with the mean value over a time period.

While previous research has found a correlation between wrist muscle activity and postural control during a handstand (Clement & Rezzette, 1985), muscle activity at the elbow, shoulder and hip joints has not been investigated. The contributions of these muscle groups is likely to be important as, in comparison to upright standing, postural control of a handstand involves more than a single strategy at the wrist joint (Blenkinsop et al., 2016; Kerwin & Trewartha, 2001; Yeadon & Trewartha, 2003). It was hypothesized that muscle activity at the wrist would correlate with postural sway to a greater degree than activity at the elbow, shoulder and hip. It was also hypothesized that the wrist muscles would demonstrate the highest effort in maintaining balance, followed by shoulder girdle and arm muscles and hip muscles.

Gymnasts undergo continuously training from early years of their physical development (Kochanowicz & Kochanowicz, 2010; Kochanowicz, Kochanowicz, Niespodziński, Mieszkowski, & Biskup, 2015). They learn crucial gymnastic skills like the handstand and later master them so they can utilize them in other more difficult exercises or conditions (Hedbávný, Sklenářková, Hupka, & Kalichová, 2013; Prassas, 1988). As a result, their ability to control balance during both upright stance and a handstand increases along with their experience and age (Asseman et al., 2004; Croix et al., 2010; Gautier, Thouwarecq, & Larue, 2008). Busquets, Marina, Davids, and Angulo-Barroso (2016) showed that gymnasts' movement pattern variability was dependent upon both training experience and physical development. It would therefore be expected that younger, less experienced gymnasts would demonstrate different neuromuscular control patterns (inter-muscle contribution) during a handstand than those with greater experience. However, differences in muscle activity between experts and non-experts have not been evaluated. It was hypothesized that younger gymnasts would exhibit higher relative muscle activity because of their higher effort and that they would rely to a greater degree upon supporting control strategies at the shoulders and hips.

Insight into the control of the muscles involved in maintaining a stable handstand by novice and expert athletes will assist in the assessment of skill-improvement and will improve our knowledge of what is necessary to master a handstand. Therefore, the aims of this study were: 1) to investigate the relationship between muscle activity and postural control during a handstand, and 2) to compare neuromuscular control patterns between children who have trained to perform handstands on a flat, stable surface, and elite adult gymnasts who are able to perform a handstand on gymnastics rings.

2. Methods

2.1. Participants

Fifteen young and eleven adult elite male gymnasts aged 13.9 ± 0.7 and 23 ± 3 years, respectively participated in the study. Participants' body mass and height was 45.2 ± 7.7 kg, 154.9 ± 9.8 cm and 72.1 ± 3.3 kg, 172.8 ± 4.3 cm for young and adult gymnasts respectively.

The criteria for participant inclusion were: a) all gymnasts started their training career at the age of six or seven; b) all participants trained for a minimum of 22 h a week; c) elite gymnasts were considered eligible if they were able to maintain a flawless handstand on the rings for five seconds. This study was conducted with the approval of the Bioethics Committee at the Regional Medical Chamber in Gdansk (approval number KG -12/15). All participants and children's legal guardians gave written informed consent prior to their participation.



Fig. 1. Handstand performed on a force platform. Directions of the center of pressure sway in the Y axis are indicated.

2.2. Procedure

Posture and muscle activity were assessed during performance of a handstand on a force platform (Fig. 1). Participants then performed several isolated maximal voluntary isometric contractions (MVC) for EMG normalization purposes. Participants were familiarized with the study protocol prior to testing and performed a 15-minute warm-up routine at a gymnastic hall at the beginning of the examination.

2.3. Posturography

Each gymnast performed three handstands of 10-second duration on the AccuGait AMTI (Advanced Mechanical Technology Inc, USA) force platform. Ground reaction forces were recorded using AMTI software with a sampling frequency of 100 Hz. Each gymnast individually chose their preferred hand position which remained the same for all attempts by marking finger positions on the platform. Handstand execution was monitored by a national gymnastics judge and was considered valid for analysis when the performance met the criteria of International Gymnastic Federation (*Federation Internationale de Gymnastique*, 2015). If one or more handstands failed to meet the criteria participants repeated the trial until three acceptable handstands were recorded. To minimize the fatigue effect, a gymnastic coach helped to establish and stabilize the handstand position and the measurements started when the contact of their hands with the gymnast's calves ended. A one-minute rest was taken between attempts. The following postural control variables were taken for sway analysis: CoP range of displacements (cm) and peak velocity ($\text{cm}\cdot\text{s}^{-1}$) in the sagittal axis X and frontal axis Y, and the overall path length (cm) and surface area of the 95th percentile ellipse (cm^2). Furthermore, maximal CoP displacement in the Y axis (forward, backward) was also analyzed to quantify the highest amplitude of postural sway in the anterior-posterior direction. The average value of the three trials was used to be representative of each participant for further calculations.

2.4. EMG recording

Participants' skin was prepared for EMG evaluation according to SENIAM organization guidelines (Hermens et al., 2000), including scrubbing, cleaning with alcohol and shaving if necessary. Placement of Ag/AgCl bipolar surface electrodes (1 cm^2 of active area, Sorimex, Toruń, Poland) was also consistent with SENIAM and Cram's guidelines (Criswell, 2011). The following muscles ($n = 12$) were investigated: trapezius descendens (TD), latissimus dorsi (LD), anterior deltoid (AD), triceps brachii (TB), biceps brachii (BB), wrist extensors (WE), wrist flexors (WF), pectoralis major (PM; pars abdominalis), multifidus (MF), gluteus maximus (GM), rectus abdominis (RA), and rectus femoris (RF).

Signals were gathered unilaterally from the dominant side of the body (determined by preferred writing hand) using a bipolar method. The EMG signals were recorded, processed and analyzed using Noraxon's TeleMyo DTS surface electromyography system and MyoResearch 1.08 software (Noraxon, Scottsdale, AZ, USA). Raw data were recorded at a sampling frequency of 1500 Hz and amplified with a gain of 500. The common mode rejection ratio was above 100 dB and the input impedance above 100 M Ω . A first order bandpass filter of 10–500 Hz ($\pm 10\%$ Hz) was used. Data were then fully rectified and smoothed by calculating the root mean square (RMS_{EMG}) values using 50 ms time frames. To reduce inter-individual variability and to evaluate relative muscle activity, RMS_{EMG} was normalized to the signal acquired during an arbitrary angle isometric MVC (isometric- arb_{MVC}) by calculating peak values in 1 s time frames, and expressed as percentage (NRMS_{EMG}) (Burden, 2010). The purpose of isometric- arb_{MVC} was to serve as the reference contraction, not to show maximal activation capacity of the involved muscles. Due to the stochastic nature of motor unit discharges, EMG variables were calculated as the average of three handstand trials with the purpose of reducing the intra-individual variability. Mean values of RMS_{EMG} were also used to show inter-muscular contributions patterns as amplitude ratios for each upper body muscle involved in the task (Donath, Kurz, Roth, Zahner, & Faude, 2016; Kurz et al., 2012). Amplitude ratios were calculated as follows: General Equation:

$$\text{Muscle "A" (\%)} = \frac{\text{Muscle "A"} \times 100}{\text{Summation of all specific muscles of interest in the ability}}$$

Example (for wrist flexors –WF):

$$\text{WF (\%)} = \frac{\text{WF} \times 100}{(\text{WF} + \text{WE} + \text{BB} + \text{TB} + \text{PM} + \text{LD} + \text{AD} + \text{TD})}$$

Muscles of the trunk and lower body (MF, GM, RA, RF) were not included in the equation due to their very low mean RMS_{EMG} values which did not exceed 1% of total analyzed muscles activation.

For normalization purposes isometric- arb_{MVC} were performed in isolated positions on the Biodex System 4 (a, b, c, d, e) or by applying manual resistance (f, g, h). In each position, the adjacent joints were stabilized with leather straps. Participants performed three 4 s isometric- arb_{MVC} , into flexion and extension for each evaluated joint and muscle:

- glenohumeral joint in the 90° flexion (normalization for AD, PM and LD),
- glenohumeral joint in 90° abduction (normalization for TD),
- elbow joint in 90° flexion with 45° of glenohumeral joint flexion (normalization for BB and TB),
- wrist joint in neutral midway between flexion and extension with 90° elbow flexion (normalization for WF and WE),
- knee joint in 90° flexion with 90° hip joints flexion (normalization for RF),
- trunk in prone position without support of the upper body (normalization for MF),
- trunk in 45° unsupported supine position with knees and hips flexed to 90° (normalization for RA),
- neutral hip angle (0°) in prone with 90° knee flexion (normalization for GM).

Table 1
Postural control during a handstand in young and adult gymnasts.

Centre of pressure (CoP) indicators	Young gymnasts		Adult gymnasts		% Δ	p	Cohen's d
	mean	SD	mean	SD			
<i>X axis (medial–lateral)</i>							
Range (cm)	3.72	1.18	2.09	0.49	−44%	0.001 ^{***}	1.80 ^{##}
Peak velocity (cm/s)	18.38	6.87	11.57	3.49	−37%	0.004 ^{**}	1.24 ^{##}
Standard deviation (cm)	0.74	0.22	0.41	0.12	−45%	0.001 ^{***}	1.86 ^{##}
<i>Y axis (anterior–posterior)</i>							
Range (cm)	6.29	0.67	4.67	0.77	−26%	0.001 ^{***}	2.24 ^{###}
Maximal forward displacement (cm)	3.41	0.58	2.37	0.36	−31%	0.001 ^{***}	2.15 ^{###}
Maximal backward displacement (cm)	2.88	0.60	2.29	0.49	−21%	0.028 [*]	1.07 [#]
Peak velocity (cm/s)	45.26	9.10	29.20	8.10	−36%	0.009 ^{**}	1.86 ^{##}
Standard deviation (cm)	1.48	0.27	0.97	0.21	−34%	0.001 ^{***}	2.11 ^{###}
<i>Overall</i>							
Path length (cm)	118.1	20.60	70.90	17.09	−40%	0.028 [*]	2.49 ^{###}
95th percentile surface area (cm ²)	21.16	9.43	7.83	4.02	−63%	0.001 ^{***}	1.83 ^{##}

Cohen's d – magnitude of effect size: # moderate, ## large, ### very large; significant difference between young and adult gymnasts at * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

2.5. Statistical analysis

To assess differences in postural control, muscle activity and inter-muscle contributions between the two groups Student's t-Test for independent samples was used. Pearson's correlation coefficient (r) was calculated to study the relationships between peak RMS_{EMG} and CoP displacement (maximal forward, backward, and surface area of the 95th percentile ellipse) and peak velocity in the Y axis. As the study was focused on EMG evaluation of one side of the body correlation analysis was performed only in the Y axis. Such analysis was made for each group separately and outliers (exceeding three standard deviations) were excluded. Cohen's d statistic was used to calculate effect size when comparing differences between groups and when studying correlations between variables [Cohen (1988) further modified by Hopkins (2006)]. Effect size magnitude was categorized as follows: small ($d = 0.2$ – 0.59 , $r = 0.1$ – 0.29), moderate ($d = 0.6$ – 1.19 , $r = 0.3$ – 0.49), large ($d = 1.2$ – 1.99 , $r = 0.5$ – 0.69), very large ($d = 2.0$ – 3.99 , $r = 0.7$ – 0.89). Normal distribution of data was checked with Shapiro-Wilk's test. All analyses were performed with commercial software: Statistica 10 (Statsoft Inc. Tulsa, OK, USA) and the level of significance was set at $\alpha = 0.05$. Results are presented as mean and standard deviation.

3. Results

3.1. Postural control

CoP displacements in X (medial–lateral) and Y (anterior–posterior) axes are presented for each group in Table 1. In all analyzed variables young gymnasts showed significantly higher values than the adult gymnasts. The highest difference between groups was seen in range and peak velocity in the X and Y axis respectively. Moreover, when comparing groups, the surface area of the 95th percentile ellipse was 270% wider in young gymnasts than in adult gymnasts.

3.2. Relative muscle activity

Peak relative muscle activity for each group is presented in Fig. 2.

The WF demonstrated the highest mean (60%) and peak (200%) NRMS_{EMG} while the WE demonstrated a NRMS_{EMG} which was three times lower. The AD, TD and TB reached mean NRMS_{EMG} of approximately 30% while all other investigated muscles did not exceed 10%. Adult gymnasts demonstrated lower peak NRMS_{EMG} in comparison to young gymnasts across all tested shoulder girdle muscles (TD, AD, TB, BB), and also showed significantly lower NRMS_{EMG} in RF. The only muscle which demonstrated lesser activity among young gymnasts was LD which had a peak NRMS_{EMG} 260% higher in adult gymnasts.

With respect to mean NRMS_{EMG} only the TB, BB and RF showed a significant difference between young and adult gymnasts. Young gymnasts had 88% ($p = 0.023$), 150% ($p = 0.003$) and 75% ($p = 0.039$) higher values compared to adults in TB, BB and RF respectively. No significant differences were found between groups for either mean or peak NRMS_{EMG} at GM, RA, MF, WF and WE.

3.3. Inter-muscle contributions

The greatest mean muscle activation as measured by amplitude ratio was observed in TD (20–26%) and WF (25.5–28%), followed by AD (15–18%) and TB (13–16%) in both young and adult gymnasts. BB and WE activity ranged from 4 to 7% while activity of PM and LD showed a 2–3% ratio of total upper body mean RMS_{EMG}. Inter-muscle contributions in young and adult gymnasts showed

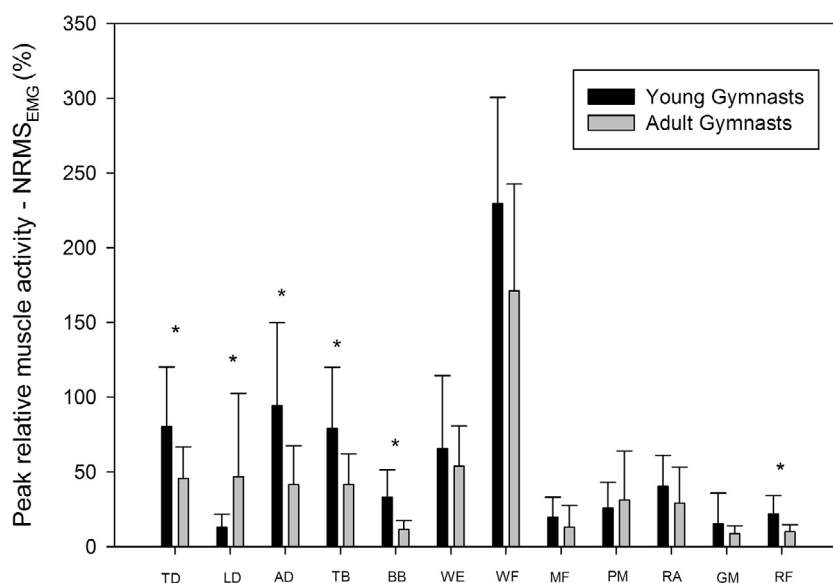


Fig. 2. Peak relative muscle EMG activity during a handstand performed by young and adult gymnasts. Note: * significant difference between young and adult gymnasts; error bars represent mean values and standard deviations. Abbreviations: TD – trapezius descendens, LD – latissimus dorsi, AD – anterior deltoid, TB – triceps brachii, BB – biceps brachii, WE – wrist extensors, WF – wrist flexors, MF – multifidus, PM – pectoralis major, GM – gluteus maximus, RF – rectus femoris.

similar patterns, although adults had a 43% higher involvement of LD than the younger gymnasts (Table 2).

3.4. Correlation analysis

Pearson's correlation coefficients for EMG activity and postural control are shown in Table 3. Young and adult gymnasts demonstrated different relationships between peak RMS_{EMG} and postural control variables. Adults' WF had significant, high and very high positive correlations for peak velocity in the Y axis, path length, surface area of the 95th percentile ellipse, in both CoP forward and backward displacements. In contrast, young gymnasts' WF peak RMS_{EMG} was very highly negatively correlated with CoP backward displacement, while WE did not show significant correlation (Table 3 and Fig. 3).

Adults' peak RMS_{EMG} of TD had a significant, highly positive correlation in both directions of the Y axis while of AD showed a high positive correlation only with the CoP maximal backward displacement. The peak velocity in the Y axis and path length was significantly and negatively correlated with the peak RMS_{EMG} of RF in young gymnasts. Maximal forward displacement and surface area of the 95th percentile ellipse of CoP were positively correlated with the peak RMS_{EMG} of RA in young gymnasts (Table 3). The peak RMS_{EMG} of GM, MF, TB, BB and WE did not show significant correlation with the analyzed postural control variables in either group.

4. Discussion

As expected, this study supports the theory that adults gymnasts are more stable during a handstand than young gymnasts and that the two groups utilize different muscle activity patterns.

The finding of greater stability among adult gymnasts is consistent with previous research (Asseman et al., 2004; Croix et al.,

Table 2

Upper body inter-muscle contributions during handstand (values represent the involvement of each muscle as a percentage of total EMG activity among all analyzed muscles).

% involvement	Young gymnasts		Adult gymnasts		% Δ	p	Cohen's d
	mean	SD	mean	SD			
Trapezius descendens (TD)	26.29	7.54	20.29	9.94	−23%	0.10	0.68 [#]
Pectoralis major (PM)	2.16	0.99	3.03	2.95	40%	0.30	0.40 [#]
Anterior deltoid (AD)	17.74	11.39	15.15	8.31	−15%	0.56	0.26 [#]
Latissimus dorsi (LD)	2.11	0.48	3.00	1.11	42%	0.01 [*]	1.04 ^{##}
Triceps brachii (TB)	13.12	3.73	16.28	8.67	24%	0.22	0.47 [#]
Biceps brachii (BB)	5.88	2.29	5.05	3.32	−14%	0.47	0.29 [#]
Wrist extensors (WE)	7.18	4.50	7.79	4.3	9%	0.74	0.14 [#]
Wrist flexors (WF)	25.51	11.99	27.75	8.52	9%	0.61	0.21 [#]

Cohen's d – magnitude of effect size: [#]small, ^{##}moderate; * significant difference between young and adult gymnasts at $p < 0.05$.

Table 3

Pearson's correlation coefficients for peak RMS_{EMG} and postural control variables in the Y axis during a handstand. Correlation analysis was done separately for young and adult gymnasts.

	CoP Max. forward displacement	CoP Max. backward displacement	Peak velocity	Path length	Surface area
Trapezius descendens (TD)	0.72 (a) [*]	0.64 (a) [*]	NS	NS	NS
Pectoralis major (PM)	NS	NS	NS	NS	NS
Anterior deltoid (AD)	NS	0.66 (a) [*]	NS	NS	NS
Latissimus dorsi (LD)	NS	NS	NS	NS	NS
Triceps brachii (TB)	NS	NS	NS	NS	NS
Biceps brachii (BB)	NS	NS	NS	NS	NS
Wrist extensors (WE)	NS	NS	NS	NS	NS
Wrist flexors (WF)	0.76 (a) [*]	0.86 (a) ^{**} ; -0.82 (y) ^{**}	0.69 (a) [*]	0.81 (a) ^{**}	0.82 (a) ^{**}
Multifidus (MF)	NS	NS	NS	NS	NS
Rectus abdominis (RA)	0.70 (y) [*]	NS	NS	NS	0.61 (y) [*]
Gluteus maximus (GM)	NS	NS	NS	NS	NS
Rectus femoris (RF)	NS	NS	-0.64 (y) [*]	0.59 (y) [*]	NS

Abbreviations: CoP – Center of Pressure.

(a) –significant correlation in adult gymnasts; (y) – significant correlation in young gymnasts; significant correlation at ^{*}p < 0.05, ^{**} p < 0.01; “NS” no significant correlation.

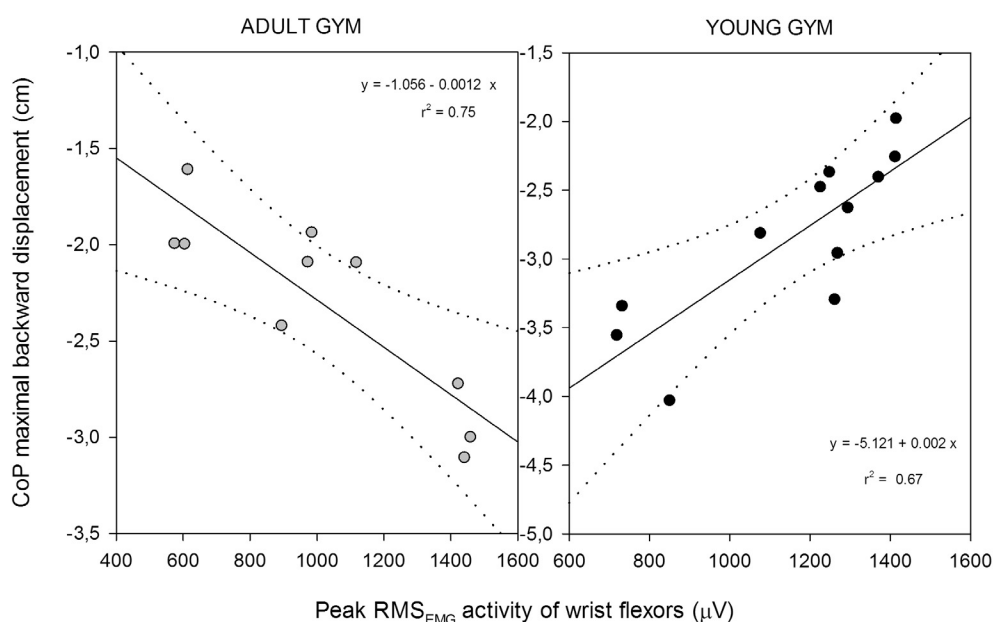


Fig. 3. Linear correlation between EMG activity and postural control variables during a handstand based on maximal backward center of pressure displacement and peak EMG activity of the wrist flexors. *Note:* Gray and black dots indicate adult and young gymnasts, respectively. Solid and dotted lines indicate linear regression and 95% confidence interval.

2010; Gautier et al., 2008) showing better postural control during handstands and upright standing among older gymnasts. It is possible that this difference is due to the effect of age rather than the effect of training, however Barozzi et al. (2014) has shown that untrained 14-year-olds perform similarly to adults during upright standing in terms of both CoP displacement and velocity. Therefore, while young gymnasts could be considered to be non-experts at handstands in comparison to adults, they are still elite athletes who have undergone long-term training. These young gymnasts demonstrate postural control performances comparable to or even beyond untrained adults (Kochanowicz, Kochanowicz, Niespodziński, Mieszkowski, & Sawicki, 2017). CoP displacements for both groups were mainly observed in the Y axis with the CoP located further forward among the young gymnasts. These displacements in the Y axis may be largely caused by limited degrees of freedom of the wrist joint as well as by higher capabilities of WF to counteract the forward sway when performing a handstand (Kerwin & Trewartha, 2001). Dynamic analysis of handstands supports the idea of the wrist joint as the primary element in maintaining balance in inverted stance (Kerwin & Trewartha, 2001; Yeadon & Trewartha, 2003). Previous authors have suggested that in order to preserve the stabilization role of the wrist joint the WF are constantly active throughout performance of a handstand (Asseman et al., 2004; Hedbávný et al., 2013; Slobounov & Newell, 1996). The high peak WF NRMS_{EMG} in comparison to other muscles and the high inter-muscle contributions ratio found in this study support the theory that the wrist joint and specifically the WF play a key role in postural stability during a handstand. It is notable that the peak NRMS_{EMG} of WF was three times greater than during isometric-arb_{MVC}. Such high values are the effect of a rapid contraction incorporating fast twitch

fibers in response to body sways mostly in forward direction (Gautier et al., 2009; Kerwin & Trewartha, 2001; Yeadon & Trewartha, 2003), as well as elongated muscle fibers during handstand position (approximately 90° wrist joint extension) in comparison to the reference contractions (Burden, 2010).

While the highest relative activity and inter-muscle contributions ratio was observed in the WF for both groups, the correlation with postural control performance in young and adult gymnasts was different. Only the adult group succeeded in using the WF as a clear counter response to anterior sways as demonstrated by the positive correlation with CoP maximal forward displacement. This suggests that in adult gymnasts the main strategy used to control balance was the wrist strategy. The same outcome was observed by Clement and Rezzette (1985) who found that well-trained experts demonstrated a forward displaced CoP compared with novices, resulting in the WF conducting the majority of postural control efforts to bring CoP in the opposite direction (backward). Clement and Rezzette (1985) also investigated postural control during handstands with simultaneous EMG recording, however their participants utilized handgrips on parallel bars rather than a flat surface. This difference makes it difficult to compare the results of muscle activation to the current study as it change the antagonistic muscle activity in wrist joint along with other muscles (Kochanowicz, Niespodziński, Marina, et al., 2017). It is also important to note that Clement and Rezzette (1985) focused on pronator teres and brachioradialis muscles which act on the elbow and distal radioulnar joints rather than on the wrist (Gilroy & MacPherson, 2016; Standing, 2008). However, as the muscle sites of pronator teres and brachioradialis are partly similar to our wrist flexor and extensors sites some reference could be done. Clement and Rezzette (1985) found that the pronator teres' EMG signal in the most experienced athlete had a strong positive correlation with forward CoP sway while the antagonistic muscle had a strong positive correlation with backward sway. Although the present study found a similar correlation for WF among the adult gymnasts the correlation for WE was poor and non-significant. It should be noted that while the present study investigated CoP maximal sway, Clement and Rezzette (1985) did not clearly describe their postural control variable, making it difficult to compare results between these studies. In the current study, adults' WF peak RMS_{EMG} was positively correlated not only with forward maximal CoP sways but also in the direction of concentric contraction (i.e. backwards) and with all other analyzed postural control variables. This could indicate the specialized role of WF to not only directly counteract forward CoP displacements but also to reduce overall body sway in the Y axis (Kerwin & Trewartha, 2001). While the CoP is maintained in a further forward direction, the backward sways are reduced (Clement & Rezzette, 1985). The correlation between maximal posterior CoP sway and WF activation may be due to the WF over-correcting for forward sway. When forward sway increases the WF develop an additional torque to counteract it (positive correlation with maximal forward displacement). As the central nervous system is unable to produce the exact amount of torque required for CoM stabilization the additional torque is slightly too high, resulting in a backward sway (the highest observed in experts).

In young gymnasts WF activation was negatively correlated only with CoP maximal backward displacement. This finding may be due to the fact that young (less experienced) gymnasts tend to utilize shoulder and hip strategies to control balance in addition to a wrist strategy (Blenkinsop et al., 2017). Blenkinsop et al. (2017) described that the wrist strategy is characterized by the shoulder joint and hip joints producing torque in the same direction as at the wrist, while the shoulder and hip strategies involve torque-production in the opposite direction to the wrist joints. It was shown by Kerwin and Trewartha (2001) that when the hip or shoulder joint produces torque in the opposite direction to the wrist, wrist torque and thus muscle activity decreases. It is possible that in the present study when the forward postural sway was mild, the young gymnasts primarily utilized the wrist strategy and the amplitude of the backward sway was therefore small. However, when the forward postural sway was larger the hip and shoulder strategies complemented the wrist strategy, resulting in a greater amplitude of posterior CoP displacement and a concomitant reduction in WF activation (negative correlation).

In contrast to WF, WE relative activity in both groups did not exceed the activity level of the other studied muscles responsible for preventing a backward shift in CoP (AD, BB). Inter-muscle contributions analysis showed that WE involvement in overall muscle activity was one of the lowest and its correlation with postural control variables was inconsistent and poor. This indicates that during a handstand on a flat surface the WF rather than WE are responsible for controlling inverted posture at the wrist joints.

The shoulder and arm muscles also play an important role during a handstand. The overall high relative EMG activity of the shoulder girdle and arm muscles during a handstand is most likely a result of their coactivation with the purpose of providing sufficient stiffness to the joints to reduce postural sway (Reynolds, 2010). After the wrist, shoulder and hip strategies play a major role in postural control during a handstand (Kerwin & Trewartha, 2001; Yeadon & Trewartha, 2003). The more experienced the gymnasts are the less they use shoulder and hip strategies due to better balance control gained through long-term training (Asseman et al., 2004; Croix et al., 2010; Vuillerme et al., 2001). The current study supports these ideas with young gymnasts demonstrating significantly higher relative peak EMG activity at the shoulder girdle and arm muscles in comparison to adults. However, only the adult group demonstrated a positive correlation between AD and TD peak RMS_{EMG} and postural control performance. As seen at the WF, greater activity at both muscles was associated with higher CoP displacements in the opposite direction of concentric contraction, indicating its effectiveness. This outcome is consistent with the hypothesis that the wrist joints function as the primary balance control strategy due to the fact that both joints produce torques in the same direction. Moreover, TD peak RMS_{EMG} was positively correlated with CoP maximal forward displacement, suggesting it plays a role not only in responding to sway but also reducing postural sway making the handstand more stable.

Unlike most of the investigated muscles, LD proved to have higher relative activity in adult gymnasts in comparison to young gymnasts (Fig. 2). This could be due to an increase in the advantage of the extensors in the agonist/antagonist torque ratio of the glenohumeral joint which occurs following long-term gymnastic training (Kochanowicz, Niespodziński, Mieszkowski, Kochanowicz, & Sawczyn, 2017; Mayer et al., 2001). This advantage was noted in the current study during isometric-arb_{MVC} testing. All other tested upper body muscles in adult gymnasts demonstrated less relative activity compared with young gymnasts. This finding may be due to long-term training effects resulting in both a weaker response of the adapted neuromuscular system to the same amount of

submaximal load and an increase in motor unit recruitment and synchronization (Felici, 2006; Gabriel, Kamen, & Frost, 2006).

Young gymnasts were observed to have higher relative activity of RF than adults and were the only group to demonstrate a correlation between RF activity and postural control variables, indicating greater utilization of the hip joint strategy in comparison to adults. As the higher RF EMG activity was associated with lower values of peak velocity in the Y axis and higher path length, it is likely that RF plays a role in reducing postural sway during a handstand among young gymnasts. Furthermore, the positive correlation between EMG activity, CoP maximal forward displacement and surface area of the 95th percentile ellipse observed in young gymnasts suggests a synergistic interaction between RA with RF. This rationale is supported by the fact that RA and RF most likely acted similarly to WF to counteract forward sway as all three muscles move the body in the same direction. The other tested lower body muscles (GM and MF) showed no differences in relative EMG activity between groups and no correlation with postural control. This could be due to the fact that their contractions in a handstand position extends the lower spine and hip in a forward direction which was already naturally achieved by the anti-gravitational response to inverted stance (Asseman & Gahery, 2005). Postural control in the lower body therefore appears to be maintained primarily through the activity of the hip and lower spine flexors (RF and RA). In their qualitative kinematic analysis of handstands Slobounov and Newell (1996) also observed flexion and extension at the elbow joint in addition to the previously discussed flexion and extension of the hip. In contrast, the current study did not find a correlation between TB and BB activity and postural control. Nevertheless, young gymnasts had significantly higher relative muscle activity of these muscles in comparison to adults, which may imply greater use of an elbow compensatory movement strategy during a handstand.

The observed differences in muscle activation during a handstand could be related to age rather than to training, as recruitment of motor units during maximal effort increases with physical development, particularly via the utilization of a greater number of fast twitch fibers (Dotan et al., 2012). However, the present study investigated a motor skill for which proper execution requires many years of systematic training, especially in the early years of motor development. Therefore, finding age-matched controls able to perform a handstand was not feasible. Future studies including reference activities would allow for the comparison of gymnasts to controls and may make it possible to separate the effect of age from the effect of training on neuromuscular performance. Such additional measurements for all 12 investigated muscle sites was beyond the scope of this study as the main focus was to evaluate overall EMG activity patterns during a handstand. However, it has been shown that the kinetics and muscle activity of young gymnasts is similar to untrained adults (Dotan et al., 2013). Therefore, the findings of this study are likely to be largely related to the gymnasts' experience.

Depending on the study's outcome and available literature, certain recommendation can be given to young handstand performers to improve their skill. While a forward location of the CoP is associated with less body sway in expert gymnasts, novices should increase the muscle strength of the shoulder extensors like LD, to be able to stabilize a forwarded position of the center of mass.

The purpose of this study was to evaluate muscle activity during a static handstand. The study was limited by a lack of motion analysis, a factor which would complement EMG and dynamic variables and enhance our understanding of the mechanics of the handstand in gymnastics and acrobatic sports.

5. Conclusions

Adult gymnasts demonstrated better postural control performance than younger gymnasts in terms of center of pressure displacement associated with lower relative muscle activity of the investigated muscles. Despite comparable muscle contributions in both groups during a handstand, expert and novice gymnasts demonstrated different relationship between muscle activity and postural control variables. In both groups, the highest relative activity of the wrist flexors, confirmed the utilization of these muscles as a primary strategy for maintaining balance during a handstand. While adults-experts rely more on the wrist flexors along with the anterior deltoid to minimize the CoP sway, young novice gymnasts who rely preferably on wrist and shoulder strategies were less effective, and therefore need a complementary activation of lower body muscles.

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Conflict of interest

The authors have no conflict of interest.

References

- Asseman, F., Caron, O., & Crémieux, J. (2004). Is there a transfer of postural ability from specific to unspecific postures in elite gymnasts? *Neuroscience Letters*, 358, 83–86.
- Asseman, F., & Gahery, Y. (2005). Effect of head position and visual condition on balance control in inverted stance. *Neuroscience Letters*, 375, 134–137.
- Bairstow, P. J., & Laszlo, J. I. (1981). Kinaesthetic sensitivity to passive movements and its relationship to motor development and motor control. *Developmental Medicine & Child Neurology*, 23, 606–616.
- Barozzi, S., Soggi, M., Soi, D., Di Berardino, F., Fabio, G., Forti, S., ... Cesarani, A. (2014). Reliability of postural control measures in children and young adolescents.

- European Archives of Oto-Rhino-Laryngology, 271, 2069–2077.
- Blenkinsop, G. M., Pain, M. T., & Hiley, M. J. (2016). Evaluating feedback time delay during perturbed and unperturbed balance in handstand. *Human Movement Science*, 48, 112–120.
- Blenkinsop, G. M., Pain, M. T. G., & Hiley, M. J. (2017). Balance control strategies during perturbed and unperturbed balance in standing and handstand. *Royal Society Open Science*, 4, 161018.
- Borg, F., Finell, M., Hakala, I., & Herrala, M. (2007). Analyzing gastrocnemius EMG-activity and sway data from quiet and perturbed standing. *Journal of Electromyography and Kinesiology*, 17, 622–634.
- Burden, A. (2010). How should we normalize electromyograms obtained from healthy participants? What we have learned from over 25 years of research. *Journal of Electromyography and Kinesiology*, 10, 1023–1035.
- Busquets, A., Marina, M., Davids, K., & Angulo-Barroso, R. (2016). Differing roles of functional movement variability as experience increases in gymnastics. *Journal of Sports Science and Medicine*, 15, 268–276.
- Clement, G., & Rezzette, D. (1985). Motor behavior underlying the control of an upside-down vertical posture. *Experimental Brain Research*, 59, 478–484.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). New Jersey: Lawrence Erlbaum.
- Criswell, E. (2011). *Cram's introduction to surface electromyography* (2nd ed.). Sudbury MA: Jones and Bartlett Publishers.
- Croix, G., Chollet, D., & Thouvaecq, R. (2010). Effect of expertise level on the perceptual characteristics of gymnasts. *The Journal of Strength & Conditioning Research*, 24, 1458–1463.
- Daly, R. M. (2010). Independent and combined effects of exercise and vitamin D on muscle morphology, function and falls in the elderly. *Nutrients*, 2, 1005–1017.
- Donath, L., Kurz, E., Roth, R., Zahner, L., & Faude, O. (2016). Leg and trunk muscle coordination and postural sway during increasingly difficult standing balance tasks in young and older adults. *Maturitas*, 91, 60–68.
- Dotan, R., Mitchell, C. J., Cohen, R., Gabriel, D., Klentrou, P., & Falk, B. (2013). Explosive sport training and torque kinetics in children. *Applied Physiology, Nutrition, and Metabolism*, 38, 740–745.
- Dotan, R., Mitchell, C., Cohen, R., Klentrou, P., Gabriel, D., & Falk, B. (2012). Child-adult differences in muscle activation—a review. *Pediatric Exercise Science*, 24, 2–21.
- Federation Internationale de Gymnastique, F. (2015). Code of Points. Men's Artistic Gymnastics. In Moutier.
- Felici, F. (2006). Neuromuscular responses to exercise investigated through surface EMG. *Journal of Electromyography and Kinesiology*, 16, 578–585.
- Gabriel, D. A., Kamen, G., & Frost, G. (2006). Neural adaptations to resistive exercise: Mechanisms and recommendations for training practices. *Sports Medicine*, 36, 133–149.
- Gautier, G., Marin, L., Leroy, D., & Thouvaecq, R. (2009). Dynamics of expertise level: Coordination in handstand. *Human Movement Science*, 28, 129–140.
- Gautier, G., Thouvaecq, R., & Chollet, D. (2007). Visual and postural control of an arbitrary posture: The handstand. *Journal of Sports Sciences*, 25, 1271–1278.
- Gautier, G., Thouvaecq, R., & Larue, J. (2008). Influence of experience on postural control: effect of expertise in gymnastics. *Journal of Motor Behavior*, 40, 400–408.
- Gilroy, A. M., & MacPherson, B. R. (2016). *Atlas of anatomy* (3rd ed.). Thieme Medical Publishers pp. 760.
- Hedbávný, P., Sklenářková, J., Hupka, D., & Kalichová, M. (2013). Balancing handstand on the floor. *Science of Gymnastics Journal*, 5, 69–80.
- Hermens, H. J., Freriks, B., Merletti, R., Stegeman, D., Blok, J., Rau, G., ... Hägg, G. M. (2000). *European recommendations for surface electromyography: results of the SENIAM project*. Enschede: Roessingh Research and Development.
- Hopkins, W. G. (2006). A scale of magnitudes for effect statistics. In *A New View of Statistics* (Internet Society for Sport Science). Available from: <http://www.sportsci.org/resource/stats/effectmag.html>.
- Horak, F. B., & Macpherson, J. M. (2011). Postural orientation and equilibrium. *Comprehensive Physiology*, 255–292.
- Houdijk, H., Brown, S. E., & van Dieën, J. H. (2015). Relation between postural sway magnitude and metabolic energy cost during upright standing on a compliant surface. *Journal of Applied Physiology*, 119, 696–703.
- Johnson, T. K., & Woollacott, M. H. (2011). Neuromuscular responses to platform perturbations in power- versus endurance-trained athletes. *Perceptual and Motor Skills*, 112, 3–20.
- Kerwin, D. G., & Trewartha, G. (2001). Strategies for maintaining a handstand in the anterior-posterior direction. *Medicine and Science in Sports and Exercise*, 33, 1182–1188.
- Kochanowicz, K., & Kochanowicz, A. (2010). Criteria of initial selection of candidates for artistic gymnastics. *Baltic Journal of Health and Physical Activity*, 2, 75–84.
- Kochanowicz, A., Kochanowicz, K., Niespodziński, B., Mieszkowski, J., & Biskup, L. (2015). The level of body balance in a handstand and the effectiveness of sports training in gymnastics. *Baltic Journal of Health and Physical Activity*, 7, 117–124.
- Kochanowicz, A., Kochanowicz, K., Niespodziński, B., Mieszkowski, J., & Sawicki, P. (2017). Effects of systematic gymnastic training on postural control in young and adult men. *Science of Gymnastics Journal*, 9, 5–15.
- Kochanowicz, A., Niespodziński, B., Marina, M., Mieszkowski, J., Kochanowicz, K., & Zasada, M. (2017). Changes in the muscle activity of gymnasts during a handstand on various apparatus. *The Journal of Strength & Conditioning Research*.
- Kochanowicz, A., Niespodziński, B., Mieszkowski, J., Kochanowicz, K., & Sawczyn, S. (2017). The effect of gymnastic training on muscle strength and co-activation during isometric elbow and glenohumeral flexion/extension. *The Journal of Sports Medicine and Physical Fitness*.
- Kurz, E., Anders, C., Herbsleb, M., Puta, C., Czepa, D., & Hilberg, T. (2012). Ankle muscle activation in people with haemophilia. *Haemophilia*, 18, 948–954.
- Massion, J. (1994). Postural control system. *Current Opinion in Neurobiology*, 4, 877–887.
- Mayer, F., Axmann, D., Horstmann, T., Martini, F., Fritz, J., & Dickhuth, H. H. (2001). Reciprocal strength ratio in shoulder abduction/adduction in sports and daily living. *Medicine and Science in Sports and Exercise*, 33, 1765–1769.
- Prassas, S. G. (1988). Biomechanical model of the press handstand in gymnastics. *International Journal of Sport Biomechanics*, 4, 326–341.
- Reynolds, R. F. (2010). The ability to voluntarily control sway reflects the difficulty of the standing task. *Gait & Posture*, 31, 78–81.
- Shvartz, E. (1968). Energy cost of a handstand. *The Journal of Sports Medicine and Physical Fitness*, 8, 18–20.
- Slobounov, S., & Newell, K. (1996). Postural dynamics in upright and inverted stances. *Journal of Applied Biomechanics*, 12, 185–196.
- Sobera, M., Siedlecka, B., Piestrak, P., Sojka-Krawiec, K., & Graczykowska, B. (2007). Maintaining body balance in extreme positions. *Biology of Sport*, 24, 81–88.
- Standing, S. (2008). Pectoral girdle and upper limb. In (40th ed.). D. Johnson (Ed.). *Gray's anatomy – The anatomical basis of clinical practice* (pp. 1576–). Churchill Livingstone: Elsevier.
- Vuillerme, N., Danion, F., Marin, L., Boyadjian, A., Prieur, J. M., Weise, L., & Nougier, V. (2001). The effect of expertise in gymnastics on postural control. *Neuroscience Letters*, 303, 83–86.
- Winter, D. A. (1995). Human balance and posture control during standing and walking. *Gait & Posture*, 3, 193–214.
- Yeadon, M. R., Rosamond, E. L., & Hiley, M. J. (2012). The biomechanical design of a gymnastics training aid for a handstand on the rings. *Proceedings of the Institution of Mechanical Engineers, Part P: Journal of Sports Engineering and Technology*, 226, 24–31.
- Yeadon, M. R., & Trewartha, G. (2003). Control strategy for a hand balance. *Motor Control*, 7, 411–430.